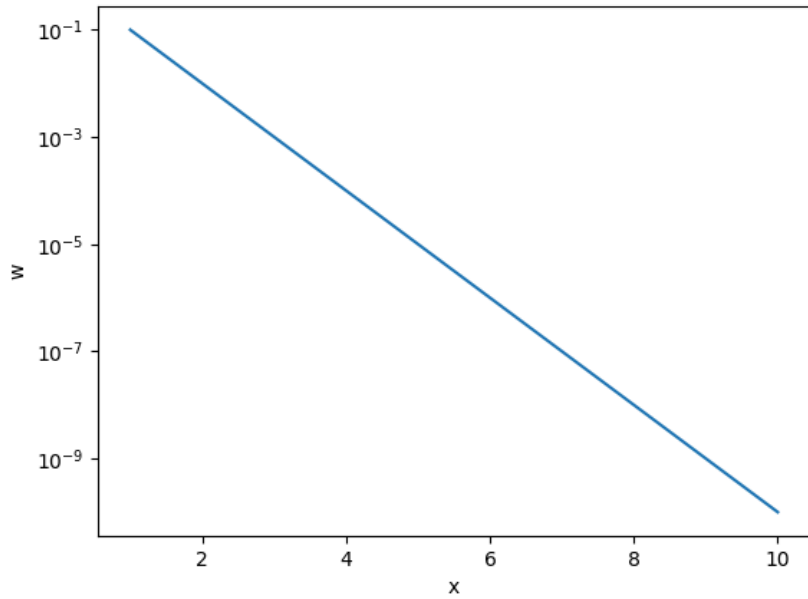
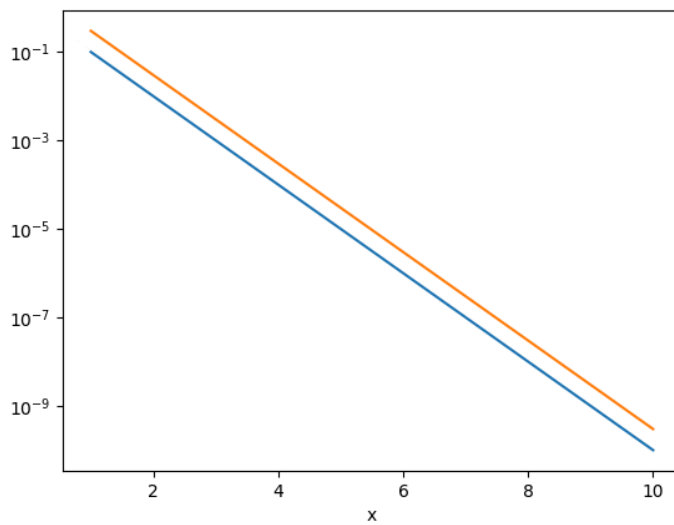


3.2

- 1) `x = np.linspace(1,10,10), y = np.arange(1,11,1)`
- 2) `x[0], x[1], x[2]`
- 3) `print('the first three entries of x are', x[0], x[1], x[2])`
- 4) $10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}, 10^{-7}, 10^{-8}, 10^{-9}, 10^{-10}$



5)



4.2

1) Using $\cos(x)$ and $\sin(x)$, I got a really small number e^{-16} so I assume this is bc of machine epsilon and sufficient for being 0?

2)

```
testDot.py •
testDot.py > dotProduct
26 import numpy as np
27 import numpy.linalg as la
28 import math
29
30 def driver():
31
32     n = 2
33     #x = np.linspace(0,np.pi,n)
34
35     # this is a function handle. You can use it to define
36     # functions instead of using a subroutine like you
37     # have to in a true low level language.
38
39     mat1 = [[1,0],[0,1]]
40     mat2 = [[1,2],[3,4]]
41
42     y = mat1
43     w = mat2
44
45     # evaluate the dot product of y and w
46     dp = dotProduct(y,w,n)
47
48     # print the output
49     print('the dot product is : ', dp)
50     return
51
52 def dotProduct(x,y,n):
53     # Computes the dot product of the n x 1 vectors x and y
54     dp = np.zeros((n,n))
55     for i in range(n):
56         for j in range(n):
57             for k in range(n):
58                 dp[i][j] += x[i][k]*y[k][j]
59     return dp
60
61 driver()
```

3) You can do `np.dot(x,y)`, numpy is better